



control units



mindy TT1N

Instructions and warnings for the fitter

Istruzioni ed avvertenze per l'installatore

Instructions et recommandations pour l'installateur

Anweisungen und Hinweise für den Installateur

Instrucciones y advertencias para el instalador

Instrukcje i uwagi dla instalatora

Aanwijzingen en aanbevelingen voor de installateur



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Warnings

The TT1 control unit is suitable for the control of a single-phase asynchronous motor powered by the electrical mains and used for the automation of awnings, rolling shutters and similar fixtures.

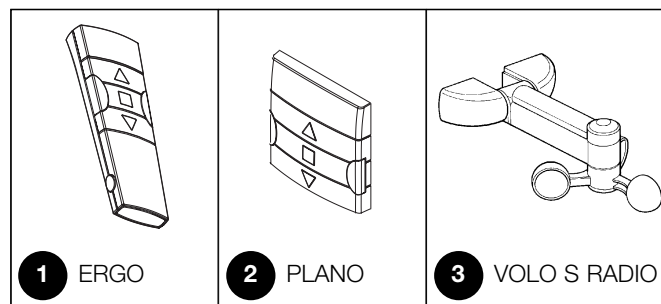
Any other use is improper and prohibited. The unit must be installed by qualified technicians in compliance with the electrical and safety regulations in force.

1) Product description

The TT1 control unit enables the control of a single-phase asynchronous motor powered by the electrical mains, with "COMMON" "UP" "DOWN" type connections, and used for the automation of awnings, rolling shutters and similar fixtures.

The control unit incorporates a radio receiver operating at a frequency of 433.92 MHz, with rolling code technology that guarantees high levels of security. Up to 30 transmitters ("ERGO" fig. 1, "PLANO" fig. 2) or radio-controlled sensors ("VOLO S RADIO" fig. 3) can be memorized for each control unit. After each command, the motor is powered for the operating time required. An electric limit switch incorporated in the motor stops the movement when the desired position is reached. The programming can be done directly from the transmitters, with beeps that sound to guide users through the various phases. "VOLO S RADIO" climatic sensors (Fig. 3) enable users to manage the movement of the awnings or rolling shutters automatically, depending on the wind, sun or rain conditions.

N.B.: Besides "ERGO", "PLANO" and "VOLO S RADIO", the control unit can manage other kinds of transmitters and functioning modes, too. Please refer to Chapter 4 "Additional information" for further explanations.



2) Installation

⚠ The electrical systems and automations must be installed by qualified and experienced personnel in compliance with current legislation. Before you proceed to make any connections make sure that the power supply is disconnected.

1. Strip the motor cable and the power supply cable about 3 cm and then the single wires approx. 5 mm.
2. Open the container by removing the "cable cap" as shown in fig. 5.
3. Thread the two cables through the designated holes in the "cable cap" (see fig. 6).
4. Pull the card a few centimetres out of the container (see fig. 7).
5. Connect the wires to the terminals as shown in fig. 8, observing the diagram in fig. 4 and the operations described in chapter 2.1.

6. Fold the cables as shown in fig. 9.
7. Push the card inside the container, make sure that the stripped length of the cable is fully inside the container, then slide the "cap" until the container closes completely (see fig. 10)
8. The control unit can be fitted directly in the shutter/awning box, double-sided tape can be used for this. To avoid the risk of water leaks it should be positioned with the cables towards the bottom as shown in Fig. 11. Do not place it with the cables towards the top (Fig. 12).

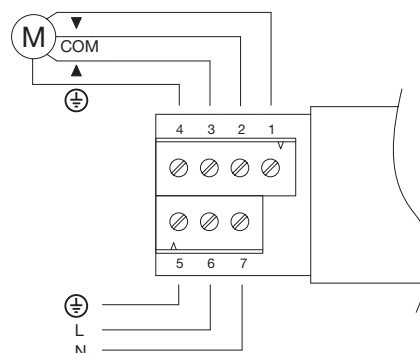
⚠ Do not perforate the container.

2.1) Electrical connections

⚠ Carefully follow all the connection instructions, if you have any doubts do NOT make experiments but consult the relevant technical specifications sheets which are also available on the web site www.niceforyou.com

An incorrect connection may cause serious damage to the control unit.

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2.1.1) Motor connection

The single-phase asynchronous motor powered by the electrical mains must be connected on terminals 1-2-3-4 (DOWN, COMMON, UP, EARTH).

"DOWN" is controlled by the ▼ key on the remote control, and "UP" by the ▲ key. (anemometer intervention direction).

If the rotation direction is not correct, invert the terminals 1 and 3.

⚠ do not connect more than one motor to each control unit; if necessary, use the appropriate expansion cards.

2.1.2) Power supply

The main supply to the unit must be connected using the terminals 5-6-7 (earth, phase, neutral) as shown in Fig. 4

2.1.3) Weather sensors

The control unit manages the climatic sensors by means of a "VOLO S RADIO"-type radio (max. no. 3). Memorisation of a "VOLO S RADIO" sensor must be carried out like that of a normal transmitter. Follow the procedure set out in Table "A2". Intervention levels must be programmed actually on the "VOLO S RADIO" sensor.

Wind intervention levels must be given priority, followed by the rain and sun. Please refer to the "VOLO S RADIO" manual for further details.

⚠ interventions on the anemometer provoke a command equivalent to the ▲ key command on the transmitters

3) Programming

Each transmitter or radio sensor is recognised by the control unit by means of an unequivocal "code". A "memorisation" phase must therefore be performed in order to allow the control unit to recognise each single transmitter.

⚠ • All the memorisation sequences are timed, that is, they must be completed within the programmed time limits.

- **For transmitters with multiple "groups", choose the group the motor must be associated with before proceeding with the memorisation phase.**
- **Programming via radio may be done on all the control units within the range of the transmitter; therefore, only the one involved in the operation should be kept switched on.**

When the memory contains no codes the first radio control unit can be entered as follows:

Table "A1"	Memorizing the first transmitter (fig. 13)	Example
1.	As soon as the control unit is powered, 2 long beeps will sound	 
2.	Within 5 seconds press and hold down button ■ of the transmitter to be memorised (for approx. 3 seconds)	 3s
3.	Release button ■ when you hear the first of the 3 beeps confirming memorisation	

N.B. If transmitters have already been memorised in the control unit, 2 short beeps will be heard when it is switched on. This means that the above procedure is not valid and another memorisation procedure must be used (Table "A2").

When one or more transmitters have already been memorised, others may be enabled as follows:

Table "A2"	Memorizing the first transmitter (fig. 14)	Example
1.	Press and hold down button ■ of the new transmitter until you hear a beep (after about 5 seconds) then release it	New  5s
2.	Press button ■ of a previously memorized transmitter slowly 3 times	Old  X3
3.	Press button ■ on the new transmitter again, releasing it when you hear the first of the 3 beeps.	New  

N.B. 3 long beeps will sound if memorisation has been successfully completed. 6 beeps will sound when the memory is full (30 transmitters), telling you the transmitter cannot be memorised.

If you need to delete all the data contained in the memory of the control unit, carry out the following procedure.

The memory can be deleted:

- using a non-memorised transmitter starting from point A.
- using a previously memorised transmitter starting the procedure from point N. 1

You can delete:

- the transmitters only, by stopping at point 4 of the procedure;
- all the data (transmitters and programming the working time), by carrying out the procedure through to point 5.

Table "A3"	Memory deletion (fig. 15)	Example
A	Switch the control unit off and cut the jumper on the weld side of the card (see fig. 15).	 
B	Power the control unit and wait for the initial beeps	 
1	Press and hold down button ■ of a previously memorized transmitter until you hear a beep (after about 5 seconds) then release it.	 5s
2	Press and hold down button ▲ of the transmitter until you hear 3 beeps; release button ▲ exactly during the third beep .	 
3	TPress and hold down button ■ of the transmitter until you hear 3 beeps; release button ■ exactly during the third beep .	 
4	Press and hold down button ▼ of the transmitter until you hear 3 beeps; release button ▼ exactly during the third beep .	 
5	If you wish to delete all the data in the memory, press the ▼ ▲ keys simultaneously within 2 seconds until you hear the first of 5 beeps, then release them.	 

N.B. 5 beeps will sound, indicating that all the codes in the memory have been deleted.

3.1) Programming the “Working time”

The “Working Time” is the time during which the control unit commands the motor, and both the factory time and the time after the memory has been deleted are approximately 150 seconds. If necessary, the running time can be altered from a minimum of 4 seconds to a maximum of 240. The programming procedure is carried out in the “self-recognition” state, or better, by measuring the time necessary to carry out the entire manoeuvre.

The motor must be moved and set next to a limit switch and the most difficult (and therefore slowest) manoeuvre for the motor measured. This is normally re-winding. The manufacturer recommends that users programme a working time which is a few seconds longer than the time which is strictly necessary for the manoeuvre.

Table “A4” Programming the working time (fig. 16)		Example
1.	Press and hold down button ■ of a previously memorized transmitter until you hear a beep (after about 5 seconds) then release it.	 5s
2.	Press key ■ again until you hear 4 short beeps (after approx. 5 seconds), then release	 5s
3.	Press keys ▲ (or ▼), in order to begin the manoeuvre and start up the time counting phase.	
4.	Wait for the motor to finish the manoeuvre and after a few seconds press key ■ to stop the time count. Three beeps will indicate the new working time has been memorised successfully.	
N.B. If the user wishes to reset the factory time of 150 seconds, at point 3 s/he must press key ■ until s/he hears the first of 3 beeps indicating programming has taken place.		

4) Additional Information

Besides the transmitters from the “ERGO” and “PLANO” series, the control unit will also recognise other kinds of transmitters manufactured by NICE (please refer to Chapter 4.1 “Usable transmitters”).

By using a special memorisation procedure for the transmitters, it is also possible to associate a special command to each of the transmitter keys (refer to Chapter 4.2 “Programming transmitters in Mode I and Mode II”).

4.1) Usable transmitters

The transmitters which can be used are shown in Table “A5”, together with their relative radio coding.

Table “A5”		Trasmittitori
CODIFICA		
FLOR	Rolling code	ERGO1 - ERGO4 - ERGO6
		PLANO1 - PLANO4 - PLANO6 - PLANO TIME
SMILO	Rolling code	VOLO S RADIO
		FLO1R - FLO2R - FLO4R
FLO	Fixed code	VERY VR
		SM2 - SM4
FLO	Fixed code	FLO1 - FLO2 - FLO4
		VERY VE

The transmitters have different kinds of coding, and the control unit is unable to recognise them simultaneously. The first transmitter memorised will thus determine the type of coding, and consequently the transmitters which can be memorised successively.

All codes must be deleted if the user wishes to change the type of transmitter (see tables “A3” “A10”).

The type of code can be checked by counting the number of beeps which sound when the control unit is switched on.

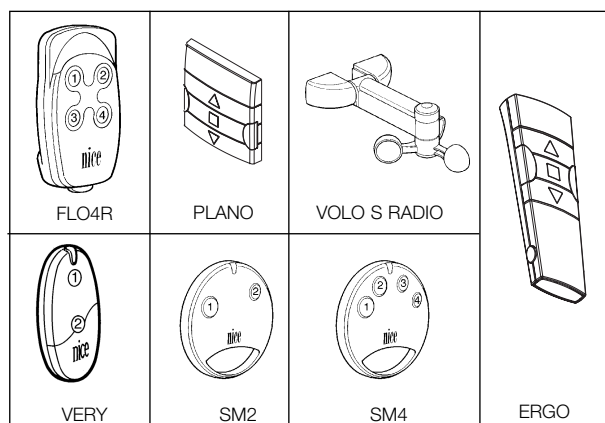


Table “A6”		Types of coding for the transmitters memorised
BIP		
1 short beep		Transmitters with FLO coding
2 short beeps		Transmitters with FLOR coding
3 short beeps		Transmitters with SMILO coding
2 long beeps		Empty memory (no memorized transmitter)

4.2) Memorizing the transmitters in Mode I and Mode II

Tables “A1” and “A2” show the memorisation of transmitters in Mode I, where each key is allocated a command:

key 1 = ▲ = ASCENT

key 2 = ■ = STOP

key 3 = ▼ = DESCENT

The transmitters can also be memorised in Mode II. This mode gives the user the most flexibility when using the transmitters.

Transmitters in both Mode I and Mode II can be memorised in the same control unit.

The differences between the 2 programming modes have been described below:

4.2.1) Mode I

In this mode, the command associated with the transmitter keys is fixed: key 1 (or ▲) commands the ascent, key 2 (or ■) commands a stop, key 3 (or ▼) commands the descent, key 4 (if present) commands a stop.

A single memorization stage is carried out for each transmitter; **during this stage it does not matter which key is pressed**, and only one space is occupied in the memory.

Esempio	memorization Mode
Key 1 or ▲	ASCENT
Key 2 or ■	STOP
Key 3 or ▼	DESCENT
Key 4	STOP

Please refer to Chapter 3 “Programming” for information about how to memorise or delete the transmitters.

4.2.2) Mode II

This mode enables the user to associate one of the following commands 1 “step step” (ascent-stop-descent-stop...), 2 “ascent”, 3 “descent”, 4 “stop” with the keys on the transmitter.

Memorisation must be carried out again if the user wishes to allocate another command to another key on the same transmitter.

The choice of which key to press is important during this phase, and space in the memory will be used up for each key memorised.

Example 1	memorization Mode II	
Key 1	ASCENT	on TT1N no. 1
Key 2	DESCENT	on TT1N no. 1
Key 3	ASCENT	on TT1N no. 2
Key 4	DESCENT	on TT1N no. 2

Example 2	memorization Mode II		
Key 1	STEP-BY-STEP on TT1N no. 1		
Key 2		STEP-BY-STEP on TT1N no. 2	
Key 3			ASCENT on TT1N no. 3
Key 4			DESCENT on TT1N no. 3



- The running time cannot be programmed if a transmitter is memorised in Mode II.
- A transmitter programmed in Mode II cannot be used in the “multi-group” mode.

When the memory contains no transmitters the first one can be memorized in mode II as follows:

Table “A7”	Memorizing the first transmitter in mode II	Example
1.	2 long beeps will sound as soon as the control unit is powered.	 
2.	Press the transmitter key to be memorised within 5 seconds, and hold it down until you hear all 3 beeps indicating memorisation has taken place, then release.	   5s
3.	Within 3 seconds press the same key of the transmitter as many times as the number corresponding to the desired command: 1 = “step-by-step” 2 = “ascent” 3 = “descent” 4 = “stop”	 1-4 3s
4.	After approximately 3 seconds, the user will hear a certain number of beeps. These beeps correspond to the command selected.	 1-4 3s
5.	Press the same key again within 2 seconds to confirm programming. Release this key when you hear the first of the 3 beeps.	  2s

N.B. 3 long beeps will sound if memorization has been successful. If, at point 4, the user does not hear the same number of beeps as the command required, wait for a few seconds to exit the procedure, without confirming the memorisation.

one or more transmitters have already been memorized, others may be enabled in Mode II as follows:

Table “A8”	Memorizing other transmitters in Mode II	Example
1.	Press and hold down the new button to be memorized until you hear a beep (after about 5 seconds) then release it	New    5s
2.	Within 5 seconds, press and hold down the key of a transmitter which has already been memorised (old) for approximately 5 seconds, until you hear 2 quick beeps, then release.	Old    5s
3.	Within 5 seconds press the same key of the (old) transmitter as many times as the number corresponding to the desired command: 1 = “step-by-step” 2 = “ascent” 3 = “descent” 4 = “stop”	Old  1-4 5s
4.	After approximately 3 seconds you will hear several beeps. The number of these beeps will be the same as the previously selected command.	 1-4 3s
5.	Press the new key of the transmitter to memorise again within 2 seconds to confirm programming. Release when you hear the first of the 3 beeps	New   2s

N.B. 3 long beeps will sound if the memorization procedure has been successful. When the memory is full (30 transmitters), 6 beeps will sound to indicate that the transmitter cannot be memorised.

A new transmitter can easily be memorised, keeping the same features as the old transmitter, by following the procedure detailed in Table "A9".

Any new transmitters which are memorised in this manner will take on the same characteristics as the old one. If the old one was memorised in Mode I, the new one will also operate in Mode I. If, on the other hand, the old one was memorised in Mode II, the key of the new transmitter will also be associated to the same command as the old one..

Table "A9"	Memorizing other transmitters (fig. 14)	Example
1.	Hold down the key of the new transmitter you wish to memorise for at least 3 seconds, then release.	New   3s
2.	Press the key of the (old) transmitter which has already been enabled for at least 3 seconds, then release.	Old   3s
3.	Press the key of the new transmitter again for at least another 3 seconds, then release.	New   3s
4.	Press the key of the old transmitter again until 3 beeps sound to confirm the new transmitter has been memorised.	Old   3s
N.B. 6 beeps will sound when the memory is full (30 transmitters) in order to indicate that the transmitter cannot be memorised.		

If you need to delete all the data contained in the memory of the control unit by using a transmitter **memorized in Mode II**, carry out the following procedure.

You can delete:

- the transmitters only, by stopping the procedure at point 4;
- all the data (transmitters and programming of the running time), by following the procedure through to point 5.

Table "A10"	Memory deletion with transmitter memorized in Mode II	Example
1.	Press and release a key memorised in Mode II. Do not worry if the motor starts up. Press the same key of the transmitter again and keep it pressed down (the motor must be off) until you hear a beep, after approximately 5 seconds, then release.	  5s
2.	Press the same key of the transmitter again and keep it pressed down until you hear 3 beeps. Release the key during the third beep.	  
3.	Press the same key of the transmitter again and keep it pressed down until you hear 3 beeps. Release the key during the third beep.	  
4.	Press the same key of the transmitter again and keep it pressed down until you hear 3 beeps. Release the key during the third beep.	  
5.	If you wish to delete all the data in the memory completely, press the same key again within 2 seconds, then release it.	  2s
N.B. After a few seconds, 5 beeps will sound to indicate that all the codes in the memory have been deleted.		

5) What to do if... a short troubleshooting guide!

When the control unit is switched on, no beep sounds and the transmitters do not transmit any commands.

Make sure that the control unit is powered correctly: there must be mains voltage between terminals 6-7. If the power supply is correct, there is likely to be a serious fault in the system, and the control unit will have to be replaced.

After a radio command, you hear 6 beeps and the manoeuvre does not start

The radio control is out of synch, the transmitter memorization procedure must be repeated.

After a radio command, you hear 10 beeps and then the manoeuvre starts

The self-diagnosis of the parameters in the memory has detected a fault. In this case, the user must delete the entire memory, memorise the remote controls and programme the running time.

You are unable to programme the running time following the procedure detailed in Table "A4"

If you are unable to programme the running time with the transmitters memorised in Mode II, check that the remote control has been inserted in the memory in Mode I (key ▲ = ASCENT, key ■ 5 = STOP, key ▼ = DESCENT)

The motor is "off", but at times you have to give the step step command twice in order to start it up.

It may be that the programmed running time is far too long for the effective duration of the manoeuvre. As the motor will have stopped next to a limit switch, the control unit may "think" the motor is still moving because of a previous command. In this case, the first command is interpreted as a STOP and the second as a movement command. Therefore, the user simply has to programme the running time correctly in order to solve the problem (refer to Chapter 3.1)

6) Technical characteristics

All technical specifications refer to a temperature of 20°C.

Electronic control unit

Power supply	: 230Vac (+10-15%) 50Hz
Maximum motor power	: 500W / 400VA
Operating temperature	: -20 ÷ 50 °C
Dimensions / weight	: 98 x 26 x 20 / 45g
Protection class	: IP54 (container undamaged)
Time taken by manoeuvre	: From 4 to 250 seconds (factory setting approx. 150 sec.)

Radio receiver

Frequency	: 433.92MHz
Coding	: FLO (fixed code), FLOR (rolling code) SMILO (rolling code)
N°. of transmitter that can be memorised	: 30 (max. 3 weather sensors)
Range of the transmitters	: estimated 150 m outdoor, 20 m indoor. *

* The capacity of the transmitters is strongly influenced by other devices with continuous transmissions which operate at the same frequency. These include alarms, headphones, etc... which interfere with the control unit receiver.

Nice S.p.a reserves the right to make changes to its products at any time as it deems necessary

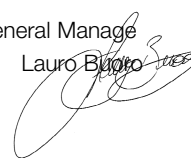
Declaration of conformity

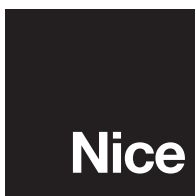
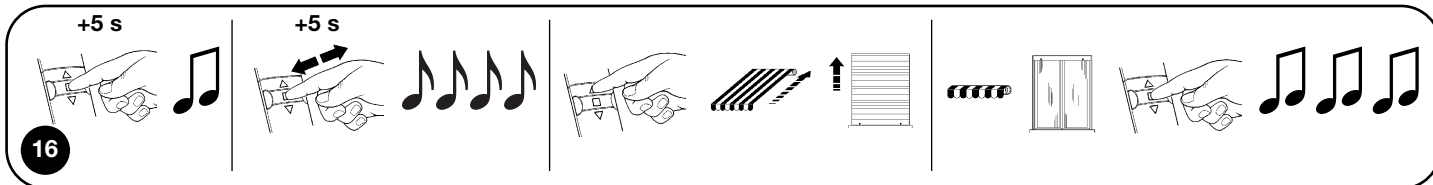
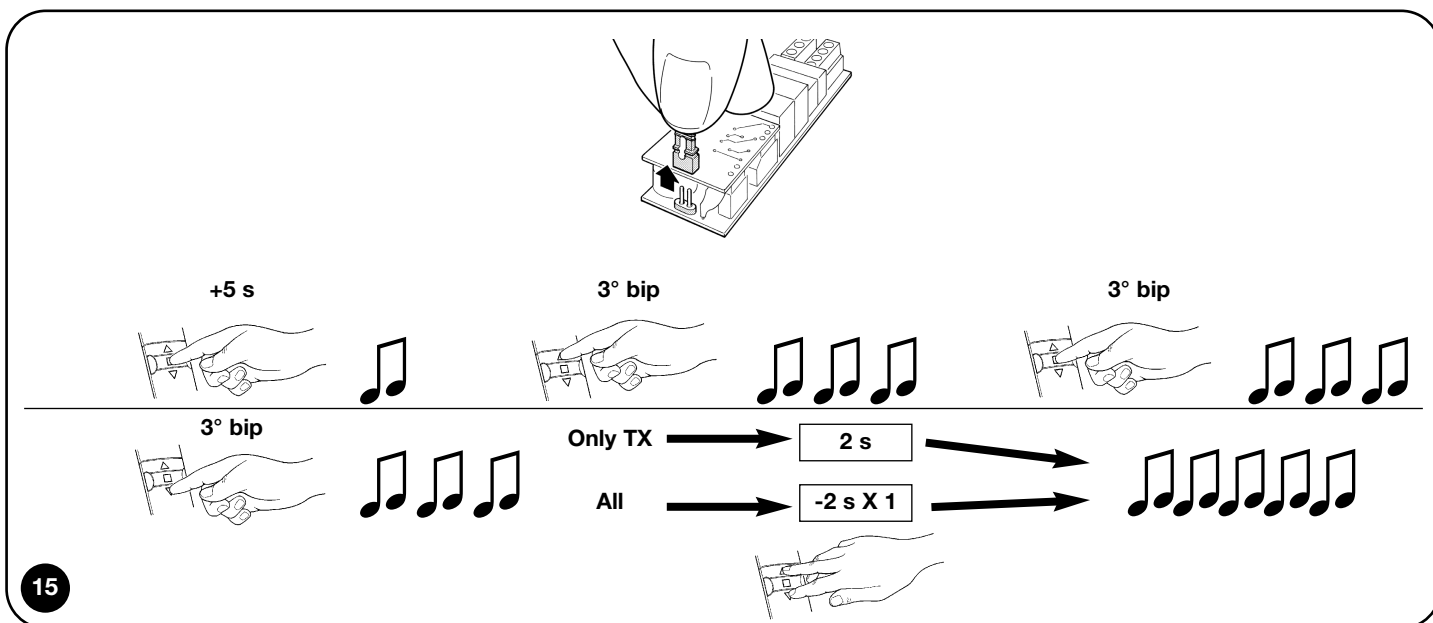
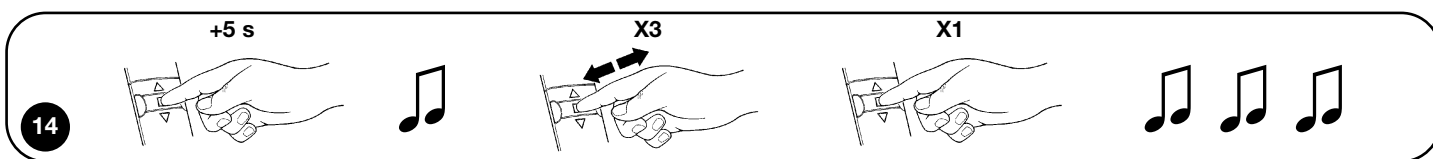
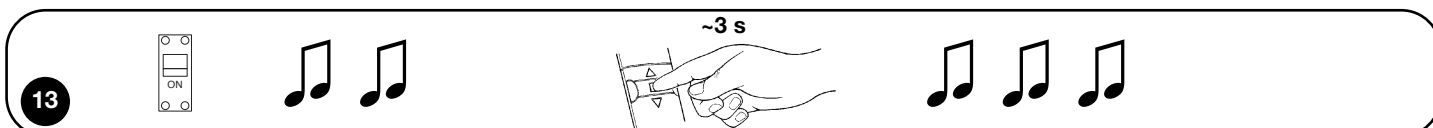
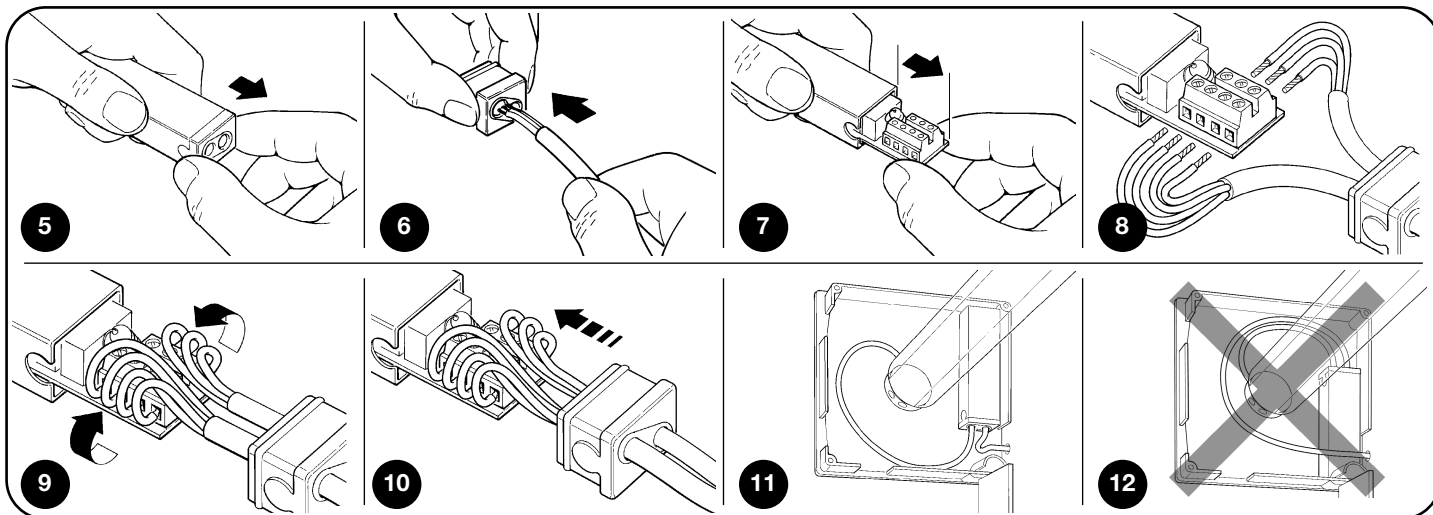
N°: mindy TT1N Rev 0

Nice S.p.a. via Pezza Alta, 13 Rustignè Oderzo (TV) ITALY declares that the product: "TT1N" complies with the essential safety provisions of Directives: R&TTE 1999/5/CE

date
20-01-2004

General Manager
Lauro Basso





Nice S.p.A.
Via Pezza Alta, 13
31046 Oderzo TV Italy
info@niceforyou.com

www.niceforyou.com